

IoT NOTICE BOARD

Project Report

Domain:Embedded Systems / Internet of Things

Controller:NodeMCU ESP8266

Communication:Wi-Fi

Display:16x2 LCD

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1. Introduction

The IoT Notice Board is an embedded-system project that demonstrates the use of a Wi-Fi-enabled microcontroller to update information on an electronic display without requiring the notice to be physically replaced. The project is intended as a simple demonstration of wireless communication, microcontroller programming and display interfacing.

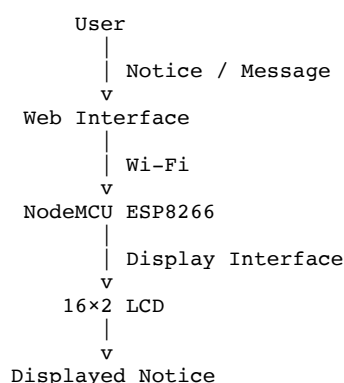
2. Objective

The main objective is to develop a notice-board system in which a message can be transmitted wirelessly and displayed electronically. The approach reduces the need for manually changing printed notices and demonstrates a practical IoT-oriented embedded application.

3. System Overview

The proposed system uses a NodeMCU ESP8266 as the central controller. Its Wi-Fi capability is used for receiving notice information, while a 16x2 LCD provides the local display output.

4. System Architecture



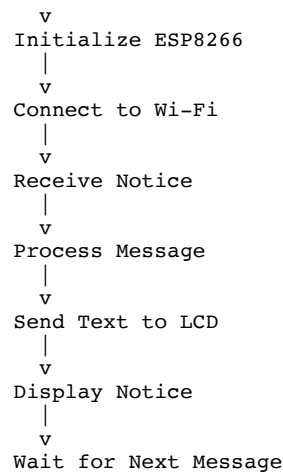
5. Hardware Components

Component	Purpose
NodeMCU ESP8266	Main controller and Wi-Fi communication
16x2 LCD	Displays the received notice
LCD interface	Connects the display to the controller
Prototype hardware	Provides electrical connections and physical assembly
Power supply	Provides power to the embedded system

6. Working Principle

The ESP8266 establishes a Wi-Fi connection and provides the communication interface through which a user can submit a notice. The received message is processed by the controller and sent to the LCD for display. The exact web-server implementation and message protocol depend on the firmware used in the original prototype.

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7. Software

The project is suitable for development using the Arduino IDE and Embedded C/C++. The firmware is responsible for initializing the ESP8266, establishing Wi-Fi communication, receiving the notice data, and controlling the LCD.

8. Key Features

- Wireless notice updating
- Wi-Fi-based communication
- ESP8266-based embedded control
- Electronic LCD display
- Remote message update concept

9. Applications

The concept can be adapted for college and school notice boards, laboratories, offices, reception areas, event announcements, public information displays and other small information-display systems.

10. Advantages

The system reduces dependence on manually replacing printed notices, provides wireless message updates, uses the built-in Wi-Fi capability of the ESP8266, and offers a relatively simple embedded implementation.

11. Limitations

The display size limits the amount of information visible at one time. Wireless operation depends on network availability, and the exact communication behavior depends on the firmware and network configuration.

12. Future Scope

Possible extensions include a mobile application, cloud-based message management, multiple display units, scrolling text, scheduled notices, message history, user authentication, real-time notifications and MQTT-based communication.

13. Project Information

Category	Details
Project	IoT Notice Board
Domain	Embedded Systems / IoT
Controller	NodeMCU ESP8266
Communication	Wi-Fi
Display	16x2 LCD
Programming	Embedded C/C++
Development	Arduino IDE
Project Type	Academic IoT Project

14. Repository Structure

```

IoT-Notice-Board/
|-- docs/
|   |-- Project-Report.pdf
|   |-- Block-Diagram.png
|   `-- Circuit-Diagram.png
|-- images/
|   |-- Prototype.jpg
|   |-- Hardware-Setup.jpg
|   `-- LCD-Display.jpg
|-- source/
|   |-- main.ino
|   `-- README.md
|-- README.md
`-- LICENSE

```

15. Conclusion

The IoT Notice Board demonstrates the integration of a Wi-Fi-enabled microcontroller and electronic display to create a remotely updateable information system. The project provides practical exposure to ESP8266 programming, wireless communication, microcontroller-based control and display interfacing.

Documentation status: This report is intended as a concise GitHub project report. It should be supplemented with the actual prototype photographs, circuit/block diagrams and verified source code when those project files are available.